

ECE 302 – Microelectronics

Description: Introduction to the physics of semiconductors, PN Junctions, BJT and MOS field Effect Transistors: Physics of operation, IV characteristics, circuit models, SPICE analysis; simple diode circuits; Single Stage Transistor Amplifiers: Common Emitter and Common Source configurations, biasing, calculations of small signal voltage gain, current gain, input resistance and output resistance;

Prerequisite: C- or better in ECE 211 — Electric Circuits

Objectives:

- Describe the essential characteristics of a semiconductor and of solid state devices.
- Explain the operation of diodes, bipolar and MOS transistors.
- Simulate circuits which contain active and passive devices with PSPICE.
- Design a single stage transistor amplifier.
- Understand basic inverter characteristics.

Lecture: 11:45AM-1PM, TuTh, 01021 EB2 Centennial Campus (Panopto recordings available)

Labs: Day and time to be announced; Please bring breadboard; Lab equipment available or use Analog Discovery 2 or 3; Parts are provided in the lab

Instructor: Dr. Shahid Ali Khan

444 MRC, Centennial Campus email: skhan29@ncsu.edu

Office hours: online via Zoom or in person; please email for appointment

TAs: TBD

Web page: Moodle – will be used for all assignments, lectures, announcements, etc.

Textbook: Microelectronic Circuit Design , Richard C. Jaeger, Travis N. Blalock, 6th edition, 2022, McGraw Hill, ebook

Sections covered (in approximate order):

Chapter 1 – Introduction to Electronics

Chapter 2 – Solid-State Electronics

Chapter 3 – Solid-State Diodes and Diode Circuits (3.1-3.7, 3.10-3.12, 3.18)

Chapter 4 – Bipolar Junction Transistors (4.1-4.9, 4.11)

Chapter 6 – Analog Systems (6.1-6.2)

Chapter 7 – Small-Signal Modeling and Linear Amplification (7.1-7.5, 7.8-7.11)

Chapter 5 – Field Effect Transistors (5.1-5.4, 5.8, 5.13-5.15; 5.6, 5.16-5.17)

Chapter 8 – Single Transistor Amplifiers (8.1-8.5, 8.7, 8.9)

Chapter 9 – Frequency Response (9.1, 9.3, 9.4.6, 9.5.1, 9.6.6)

Chapter S6 – Digital Electronics (S6.1-6.2, 6.4-6.7, 6.9)

Chapter S7 – CMOS Logic Design (S7.1-7.2, 7.5-7.6)

Grading	Homework 10%
	Laboratory 16%
	Exams (2) 49%
	Final 25%

Class policies and procedures

☐ See online Lecture Homework Schedule for details

First day of class Aug 18

Last day to drop Oct 22

Exams 09/28, 10/29

Last day of class Dec 2

Final Exam ECE 302-002: Dec 3, 11:45am-2:15pm

☐ Email and class web page will be used to make announcements. The web page has the syllabus, homework assignments, prelabs/labs, and grades. If you are registered for the class, you are automatically given access to the web page using your unity ID.

☐ Homework will be submitted online. **No late homework will be accepted.** UNITS are VERY important – please keep them in all calculations, show all work – box answers, put name/class on every sheet. Solutions for the exams will be posted on the web page. The lowest homework grade will be dropped.

☐ There are 8 weeks of labs that will begin August 27, 2025. See the online calendar for schedule and web page for prelabs and lab descriptions. Participation in labs is required if you are taking this class for credit. Late reports will receive a maximum grade of 50%.

☐ **Emergency Absences:** Students should report as soon as possible to the instructor personally but no more than one week after return to class. Examples of emergency absences are injury or illness certified by an attending physician (not the University Health Services) or death/serious illness in the family (with appropriated documentation).

☐ Reasonable accommodations will be made for students with verifiable disabilities. In order to take advantage of available accommodations, students must register with Disability Services for Students at 1900 Student Health Center, Campus Box 7509, 515-7653. NC State's policy on working with students with disabilities

☐ All the provisions of the code of academic integrity apply to this course. In addition, it is my understanding and expectation that your signature on any test or assignment means that you neither gave nor received unauthorized aid.

☐ Wellness Resources

☐ **Please contact me if you are confused about anything.**